

Surface Area of Prisms

Lesson 10-4

Name:

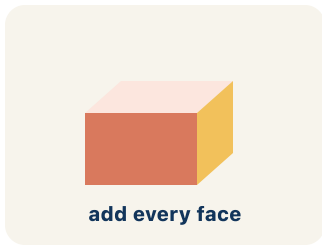
Date:

Class:

Key Vocabulary

Level 2 Standard

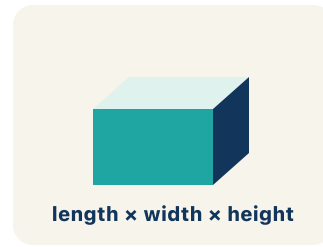
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Paint every outside surface of a box — the total painted area is the surface area

Surface area

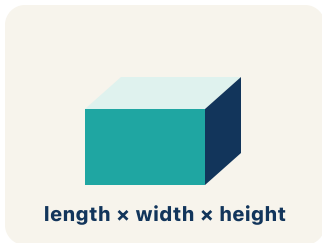
Write the definition:



A box with 6 flat rectangle sides: top, bottom, front, back, left, right

Rectangular prism

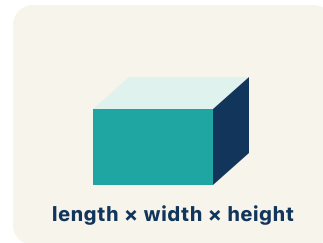
Write the definition:



Shaped like a tent or a wedge of cheese — 2 triangle ends + 3 rectangle sides

Triangular prism

Write the definition:



A triangular prism has 5 faces: 2 triangles + 3 rectangles

Face

Write the definition:

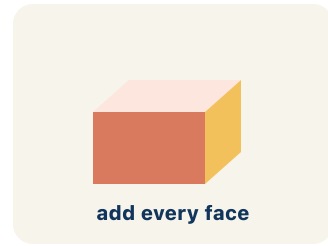


flattened faces

Unfold a triangular prism flat: you see 2 triangles and 3 rectangles side by side

Net

Write the definition:



add every face

On a triangular prism, the 3 rectangles that wrap around the sides are lateral faces

Lateral face

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the surface area of rectangular and triangular prisms.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Rectangular prism

Triangular prism

Face

Net

Lateral face

1

The total area of all the faces of a solid is its _____.

2

A solid with six rectangular faces is a _____.

3

A solid with two triangular faces and three rectangular faces is a _____.

4

A flat surface of a solid figure is a _____.

5

A flat pattern that folds into a solid is a _____.

6

A side face of a prism that is not a base is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the surface area of a rectangular prism with $l = 5$ cm, $w = 4$ cm, $h = 3$ cm?

- A. 94 cm^2
- B. 60 cm^2
- C. 94 cm^3
- D. 47 cm^2

Step 1 $SA = 2(5 \times 4) + 2(5 \times 3) + 2(4 \times 3) = 40 + 30 + 24 = 94 \text{ cm}^2$.

 **Answer:** A. 94 cm^2

 We do — together

Solve this one with your class using the same steps.

Problem: How many faces does a triangular prism have?

- A. 5
- B. 4
- C. 6
- D. 3

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: A triangular prism has two triangular bases each with area 12 cm^2 and three rectangular faces with areas 30 cm^2 , 40 cm^2 , and 40 cm^2 . What is the total surface area?

- A. 134 cm^2
- B. 110 cm^2
- C. 122 cm^2
- D. 134 cm^3

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A rectangular capsule crate is 6 cm long, 5 cm wide, and 4 cm tall. How much cardboard is needed to cover all its faces?

- A. 148 cm^2
- B. 120 cm^2
- C. 74 cm^2
- D. 148 cm^3

Show your work:

2. A wedge-shaped (triangular prism) capsule has triangle ends with base 8 in and height 6 in. The prism is 10 in long, and the triangle's three sides are 8 in, 6 in, and 10 in. How much wrapping paper covers it?

- A. 288 in^2
- B. 240 in^2
- C. 48 in^2
- D. 288 in^3

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A company ships products in two container options: a rectangular prism ($8 \times 6 \times 4$ in) or a triangular prism (triangle base 8 in, triangle height 6 in, prism length 8 in, slant sides 6.3 in each). Both must be wrapped in protective film. Which shape uses LESS film? Show your work.

Sentence starter: Rectangular prism $SA = 2(\text{ }) + 2(\text{ }) + 2(\text{ }) = \text{ } \text{ in}^2$. Triangular prism $SA = 2(\text{ }) + \text{ } + \text{ } + \text{ } = \text{ } \text{ in}^2$. The uses less film because .

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 8$ ft, $w = 6$ ft, $h = 3$ ft. What is the surface area?

- A. 180 ft^2
- B. 144 ft^2
- C. 180 ft^3
- D. 90 ft^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Triangular prism
4. **Notes 4:** Face
5. **Notes 5:** Net
6. **Notes 6:** Lateral face
7. **We Try (notes):** A. 5 — *A triangular prism has 5 faces: 2 triangular bases and 3 rectangular lateral faces.*
8. **You Try (notes):** A. 134 cm^2 — $SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 \text{ cm}^2$.
9. **Try It 1:** A. 148 cm^2 — $SA = 2(6 \times 5) + 2(6 \times 4) + 2(5 \times 4) = 60 + 48 + 40 = 148 \text{ cm}^2$.
10. **Try It 2:** A. 288 in^2 — *Two triangle ends: $2 \times (\frac{1}{2} \times 8 \times 6) = 48 \text{ in}^2$. Three rectangles: $(8 + 6 + 10) \times 10 = 24 \times 10 = 240 \text{ in}^2$. $SA = 48 + 240 = 288 \text{ in}^2$.*
11. **Exit Ticket:** A. 180 ft^2 — $SA = 2(8 \times 6) + 2(8 \times 3) + 2(6 \times 3) = 96 + 48 + 36 = 180 \text{ ft}^2$. *Surface area uses square units (ft^2).*